

Exponents and Logarithms

Rewrite each equation in exponential form.

1) $\log_{11} 121 = 2$

$$11^2 = 121$$

2) $\log_9 81 = 2$

$$9^2 = 81$$

3) $\log_7 49 = 2$

$$7^2 = 49$$

4) $\log_{216} 6 = \frac{1}{3}$

$$216^{\frac{1}{3}} = 6$$

Rewrite each equation in logarithmic form.

5) $81^{\frac{1}{2}} = 9$

$$\log_{81} 9 = \frac{1}{2}$$

6) $16^2 = 256$

$$\log_{16} 256 = 2$$

7) $7^2 = 49$

$$\log_7 49 = 2$$

8) $12^2 = 144$

$$\log_{12} 144 = 2$$

Rewrite each equation in exponential form.

9) $\log_x 191 = y$

$$x^y = 191$$

10) $\log_5 n = -2$

$$5^{-2} = n$$

11) $\log_5 x = 19$

$$5^{19} = x$$

12) $\log_n m = -6$

$$n^{-6} = m$$

Rewrite each equation in logarithmic form.

13) $x^y = 178$

$$\log_x 178 = y$$

14) $19^{-19} = x$

$$\log_{19} x = -19$$

15) $x^y = z$

$$\log_x z = y$$

16) $b^a = 154$

$$\log_b 154 = a$$